# HU Collection

Test of Mathematics for University Admission

**Mock Set C  Paper 2**

Mathematical Reasoning

| | |
|---|---|
| Time allowed: | **75 minutes** |
| Questions: | 20 multiple choice |
| Calculators: | not permitted |
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| Date written: | 23 September 2026 |

Name: _____

Date sat: _____

Score: _____ / 20

**Instructions**

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. All angles are in radians. Throughout, "must be true" means true in every case allowed by the information given. This set is weighted towards implications and their negations, redundant and independent conditions, and reasoning with trigonometric inequalities and integrals.

| Q | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans | | | | | | | | | | |
| Q | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans | | | | | | | | | | |

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**Question 1**

Consider the statement

"The train will be on time unless it snows."

Here "$A$ unless $B$" means "if not $B$, then $A$". Which one of the following is the negation of the statement?

**A** It snows and the train is on time.
**B** It snows and the train is not on time.
**C** If it snows, then the train will not be on time.
**D** If the train is not on time, then it snowed.
**E** It does not snow and the train is not on time.
**F** It does not snow or the train is not on time.

**Question 2**

$P$ and $Q$ are statements, and the statement "if $P$ then $Q$" is false. Consider:

**I** if not $P$, then $Q$
**II** if $Q$, then not $P$
**III** if $P$, then not $Q$
**IV** if not $Q$, then not $P$

Which of these must be true?

**A** I only
**B** II only
**C** I and II only
**D** I, II and III only
**E** IV only
**F** I, II, III and IV

**Question 3**

Which of the following statements are logically equivalent to "if $P$, then $Q$ or $R$"?

**I** If $P$ and not $Q$, then $R$.
**II** If not $Q$ and not $R$, then not $P$.
**III** Either "if $P$ then $Q$" is true, or "if $P$ then $R$" is true (or both).
**IV** "If $P$ then $Q$" and "if $P$ then $R$" are both true.

**A** I only
**B** I and II only
**C** I, II and III only
**D** II and III only
**E** I and III only
**F** I, II, III and IV



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**Question 4**

A function $f$ from the real numbers to the real numbers is called *increasing* if, for all real $x$ and $y$, if $x < y$ then $f(x) < f(y)$. Which one of the following is equivalent to "$f$ is not increasing"?

**A** For all real $x$ and $y$, if $x < y$ then $f(x) \geq f(y)$.
**B** There exist real $x$ and $y$ such that, if $x < y$ then $f(x) \geq f(y)$.
**C** There exist real $x$ and $y$ with $x \geq y$ and $f(x) \geq f(y)$.
**D** For all real $x$ and $y$, $x \geq y$ or $f(x) \geq f(y)$.
**E** There exist real $x$ and $y$ with $x < y$ or $f(x) \geq f(y)$.
**F** There exist real $x$ and $y$ with $x < y$ and $f(x) \geq f(y)$.

**Question 5**

A student wants to prove the following statement by contradiction:

for every real $x$, if $x^2 - 3x + 2 < 0$ then $1 < x < 2$.

Which of the following is a correct first line for such a proof?

**A** Suppose that for every real $x$, $x^2 - 3x + 2 \geq 0$ and ($x \leq 1$ or $x \geq 2$).
**B** Suppose there is a real $x$ with $x^2 - 3x + 2 < 0$ and ($x \leq 1$ or $x \geq 2$).
**C** Suppose there is a real $x$ with $x^2 - 3x + 2 < 0$ and $x \leq 1$ and $x \geq 2$.
**D** Suppose there is a real $x$ with $x^2 - 3x + 2 \geq 0$ and $1 < x < 2$.
**E** Suppose there is a real $x$ with $x^2 - 3x + 2 \geq 0$ or $1 < x < 2$.
**F** Suppose there is a real $x$ such that, if $x^2 - 3x + 2 < 0$ then ($x \leq 1$ or $x \geq 2$).

**Question 6**

Consider the statement

for all real $x$, if $\sin x > 0$ or $\cos x > 0$, then $\sin x + \cos x > 0$.

How many of the following values of $x$ are counterexamples to this statement?

$$\frac{\pi}{6}, \quad \frac{2\pi}{3}, \quad \frac{5\pi}{6}, \quad \frac{7\pi}{6}, \quad \frac{5\pi}{3}, \quad \frac{7\pi}{4}$$

**A** 1
**B** 2
**C** 3
**D** 4
**E** 5
**F** 6

**Question 7**

A student is asked to prove: *for every integer $n$, if $n^3$ is odd then $n$ is odd.* The student writes:

"Let $n$ be odd, so $n = 2k+1$ for some integer $k$. Then $n^3 = 8k^3+12k^2+6k+1 = 2(4k^3+6k^2+3k)+1$, which is odd. This proves the statement."

Which of the following is correct?

**A** The argument proves the converse. The statement is true, but this is not a proof of it.
**B** The argument is valid and proves the statement.
**C** The argument proves the contrapositive of the statement, so it is valid.
**D** The argument contains an algebraic error.
**E** The argument proves the converse, and the statement is false.



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**Question 8**

For a real number $x$, how many of the following statements are true for every real $x$?

**I** $x > 2$ if $x^2 > 4$.
**II** $x > 2$ only if $x^2 > 4$.
**III** $x^2 > 4$ if and only if $|x| > 2$.
**IV** $x^3 > 8$ only if $x > 2$.
**V** $x^2 > 4$ if $x^3 > 8$.
**VI** $x^2 > 4$ only if $x^3 > 8$.

**A** 1
**B** 2
**C** 3
**D** 4
**E** 5
**F** 6

**Question 9**

For a real number $x$, the condition

$$\sin x > \tfrac{1}{2}$$

is _____ for $\cos 2x < \frac{1}{2}$.

**A** necessary but not sufficient
**B** sufficient but not necessary
**C** necessary and sufficient
**D** neither necessary nor sufficient

**Question 10**

Which of the following statements are true for every real $x$?

**I** $\sin x + \cos x \leq \sqrt{2}$
**II** $\sin x \cos x \leq \frac{1}{2}$
**III** $\sin^4 x + \cos^4 x \geq \frac{1}{2}$
**IV** $\sin x + \cos x \geq -1$

**A** I only
**B** I and II only
**C** I, II and III only
**D** I, II and IV only
**E** II and III only
**F** I, II, III and IV



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**Question 11**

A student solves the inequality $\sin x \cos x > \cos^2 x$ for $0 \leq x < 2\pi$ as follows.

"When $\cos x = 0$, both sides are 0, so the inequality fails. Otherwise $\cos^2 x > 0$, so I can divide by it without changing the direction of the inequality, giving $\tan x > 1$. Hence the solution set is $\frac{\pi}{4} < x < \frac{\pi}{2}$ together with $\frac{5\pi}{4} < x < \frac{3\pi}{2}$."

Which of the following is correct?

**A** The final answer is correct and the argument is valid.
**B** The final answer is wrong, because dividing by $\cos^2 x$ may reverse the inequality.
**C** The final answer is wrong, because solutions with $\cos x < 0$ have been lost.
**D** The final answer is wrong, because it includes values where $\cos x = 0$.
**E** The final answer is correct, but the argument is invalid.
**F** The final answer is wrong; the solution set is $\frac{\pi}{4} < x < \frac{5\pi}{4}$.

**Question 12**

Which of the following statements are true for every $x$ with $0 < x < \dfrac{\pi}{2}$?

**I** $\sin x + \cos x > 1$
**II** $\sin 2x < 2 \sin x$
**III** $\cos 2x > \cos x$
**IV** $\tan x > \sin x$

**A** I only
**B** I and II only
**C** I, II and III only
**D** II and IV only
**E** I, II, III and IV
**F** I, II and IV only

**Question 13**

For a positive integer $n$, consider the four statements

$$(1)\ 4 \mid n, \qquad (2)\ 6 \mid n, \qquad (3)\ 12 \mid n, \qquad (4)\ 8 \mid n.$$

Let $T$ be the set of positive integers satisfying all four. A statement is called *removable* if deleting it from the list does not change the set of integers satisfying all the remaining statements. Which of the following is true?

**A** Each of the four statements is removable.
**B** Only statement (4) is not removable.
**C** Only statement (1) is not removable.
**D** Only statements (3) and (4) are not removable.
**E** No statement is removable.
**F** Only statements (2) and (4) are not removable.



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**Question 14**

For a positive integer $n$, consider

$$A:\ 4 \mid n, \qquad B:\ 6 \mid n, \qquad C:\ 8 \mid n^2, \qquad D:\ 12 \mid n^2.$$

Which one of the following is true?

**A** $A$ is equivalent to $C$, and $B$ is equivalent to $D$.
**B** $A$ is equivalent to $C$, but $B$ is not equivalent to $D$.
**C** $B$ is equivalent to $D$, but $A$ is not equivalent to $C$.
**D** Neither $A$ is equivalent to $C$, nor $B$ to $D$.
**E** $A$ is equivalent to $D$, and $B$ is equivalent to $C$.

**Question 15**

For a real number $x$, consider the five statements

$$x > 1, \qquad x^2 > 1, \qquad x^3 > 1, \qquad x > -1, \qquad x^2 > x.$$

Find the complete set of real $x$ for which exactly two of these statements are true.

**A** $x < -1$
**B** $-1 < x < 0$
**C** $x < 0$
**D** $x \leq 0$ and $x \neq -1$
**E** $x < 0$ and $x \neq -1$
**F** there is no such $x$

**Question 16**

Let $a$, $b$, $c$ be real numbers and $f(x) = ax^2 + bx + c$. Consider the conditions

$$(1)\ a > 0, \qquad (2)\ b^2 < 4ac, \qquad (3)\ c > 0, \qquad (4)\ f(x) > 0 \text{ for every real } x.$$

Which of the following must be true?

**I** Conditions (1) and (2) together imply (3).
**II** Condition (4) implies both (1) and (2).
**III** Conditions (2) and (3) together imply (1).

**A** I only
**B** II only
**C** I and II only
**D** I and III only
**E** II and III only
**F** I, II and III



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**Question 17**

For real numbers $a$ and $b$, consider the four conditions

$$(1)\ a+b>0, \qquad (2)\ ab>0, \qquad (3)\ a>0, \qquad (4)\ b>0.$$

How many of the six pairs of these conditions are such that the two conditions together are equivalent to all four conditions holding?

**A** 0
**B** 1
**C** 2
**D** 3
**E** 4
**F** 5

**Question 18**

The function $f$ is continuous and increasing on $[0,1]$. Which of the following must be true?

**I** $\displaystyle\int_0^1 f(x)\,dx \le f(1)$
**II** $\displaystyle\int_0^1 xf(x)\,dx \ge \frac{1}{2}\int_0^1 f(x)\,dx$
**III** $\displaystyle\int_0^{1/2} f(x)\,dx \le \int_{1/2}^1 f(x)\,dx$

**A** I only
**B** II only
**C** III only
**D** I and III only
**E** II and III only
**F** I, II and III

**Question 19**

The function $f$ is continuous on the real numbers and satisfies $f(-x) = -f(x)$ for every $x$. Let $a > 0$. Which of the following must be true?

**I** $\displaystyle\int_{-a}^{a} f(x)\,dx = 0$
**II** $\displaystyle\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$
**III** $\displaystyle\int_{-a}^{a} xf(x)\,dx = 0$
**IV** $\displaystyle\int_0^{2a} f(x-a)\,dx = 0$

**A** I only
**B** I and II only
**C** I and IV only
**D** I, II and IV only
**E** I, II, III and IV
**F** II and III only



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**Question 20**

For each positive integer $n$, $S(n)$ is a statement. It is known that $S(10)$ is true, and that for every positive integer $n$,

$$\text{if } S(n) \text{ is true then } S(n+3) \text{ is true,} \qquad \text{and} \qquad \text{if } S(n+2) \text{ is true then } S(n) \text{ is true.}$$

Which of the following is the set of positive integers $n$ for which $S(n)$ must be true?

**A** $\{10\}$
**B** $\{2, 4, 6, 8, 10\}$
**C** $\{2, 4, 6, 8, 10\} \cup \{13, 16, 19, \dots\}$
**D** $\{1, 2, \dots, 10\}$
**E** $\{10, 11, 12, \dots\}$
**F** all positive integers

**END OF PAPER**



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